

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276665

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

HALGRIN; Julien et al.

CATCH STRAP MECHANISM FOR AN AIRBAG MODULE

Abstract

A catch strap mechanism for an airbag module having an airbag, comprising at least one catch strap arrangement, wherein one end of a catch strap arrangement can be attached to the airbag, and a tightening device for the catch strap arrangement.

Inventors: HALGRIN; Julien (Doudeauville en Vexin, FR), GROS; Christophe (Poitiers, FR), JACQMARCQ; Thierry (Cuigy en Bray, FR), CHARPENTIER; Frederic (Le Vauroux, FR), MARCHAND; Marc (Le Vaumain, FR), HELLOT; Laurent (La Feuillie, FR), GAILHARD; Carole (Gournay en Bray, FR), LENGAIGNE; Laurent (La Chaussée, FR), LOUVET; Matthieu (Lyons la Foret, FR)

Applicant: Autoliv Development AB (Vargarda, DE)

Family ID: 85202104

Appl. No.: 18/848461

Filed (or PCT Filed): February 07, 2023

PCT No.: PCT/EP2023/052984

Foreign Application Priority Data

DE 10 2022 106 750.5

Mar. 23, 2022

Publication Classification

Int. Cl.: B60R21/2338 (20110101); B60R21/207 (20060101)

U.S. Cl.:

Background/Summary

DESCRIPTION

[0001] The present invention relates to a catch strap mechanism for an airbag module having an airbag, comprising at least one catch strap arrangement, wherein one end of a catch strap arrangement can be attached to the airbag, and a tightening device for the catch strap arrangement.

[0002] Such a catch strap mechanism can in particular be a component of an airbag device for a vehicle seat which comprises at least one airbag module having an airbag and which is suitable for installation in the vehicle seat. In this case, one end of the catch strap arrangement is attached to the airbag. The present invention also relates to a vehicle seat for a motor vehicle, comprising a backrest, a seat cushion and an airbag device of this kind.

[0003] Such a vehicle seat with such a catch strap mechanism is known, for example, from US 2019/0248322 A1. The airbag, which is arranged in particular in a lateral region of a backrest in the initial state, unfolds after it is triggered next to and/or in front of a passenger sitting on the vehicle seat. It can be provided that the catch strap arrangement guides the unfolding process at least temporarily during the unfolding of the airbag, for example when the catch strap arrangement is in a tensioned state. It is also conceivable that the catch strap arrangement is unwound from a spool during unfolding or is tensioned from a wound or folded state by the unfolding airbag.

[0004] The tightening device is provided to retract the catch strap arrangement during unfolding or at the end of the unfolding process and therefore to influence the unfolding process or to bring the airbag into a final position after unfolding. For example, a portion of the airbag located in front of the passenger after unfolding can be pulled towards the backrest by the tightening device so that the passenger is also pulled towards the backrest. From US 2019/0248322 A1, for example, it is known to wind up the catch strap arrangement by means of the tightening device through a rotary movement.

[0005] The object of the present invention is to provide an alternative to how the catch strap arrangement can be tightened.

[0006] A solution to the object is specified in independent claim 1, wherein further solutions and advantageous developments are specified in the preceding and following description, wherein individual features of the advantageous developments can be combined with one another in a technically reasonable manner.

[0007] The object is achieved in particular by a catch strap mechanism with the features mentioned at the outset, in which the tightening device comprises a tightening element that can be accelerated in a straight line and which can be accelerated in particular by means of a pyrotechnic drive.

[0008] The tightening device can, for example, comprise a tensioner tube in which a piston is arranged. For example, a gas generator can be used to introduce compressed gas into the tensioner tube, which drives the piston through the tensioner tube. The tightening element is in particular connected, for example, by means of a rigid rod or by means of a tension cable to the tightening element, which is arranged outside the tensioner tube.

[0009] In this context, it can be provided in particular that a deflection block, in particular designed as a single piece, is attached to the tensioner tube, through which the tension cable is guided. Preferably, a receptacle is formed in the deflection block in which the gas generator is arranged. Such an arrangement has in particular the advantage that the tensioner tube can extend obliquely to the linear direction of movement of the tightening element connected to the piston via the tension cable in a mounted state. It is therefore possible, for example, for the tensioner tube to extend in the longitudinal direction of the vehicle when the vehicle seat is in the mounted state in the usual

manner in a motor vehicle, while the rectilinear direction of movement of the tightening element extends transversely thereto.

[0010] Each catch strap arrangement can be made from a single catch strap or catch cord which is made in particular from a taught fabric. The catch strap arrangement can however also be made of a plurality of catch straps/catch cords connected to each other.

[0011] The tightening device is coupled to the catch strap arrangement via the tightening element in such a way that the catch strap arrangement is deflected by the tightening element at least before, after and/or during the tightening process. The portions arranged in front of and behind the tightening element in the catch strap running direction are therefore arranged at an angle to each other. For example, the catch strap arrangement can be deflected by the tightening element by more than 60° , preferably more than 90° or about 180° . The degree of deflection can be constant during the tightening process, but can also change during the tightening process.

[0012] The catch strap arrangement and the tightening element which can be accelerated in a straight line are in particular arranged relative to each other in such a way that a type of pulley is formed. The tightening element which can be accelerated in a straight line and the catch strap arrangement are arranged in particular in such a way that the end of the catch strap arrangement attached to the airbag covers a greater distance during the tightening process than the linearly accelerating tightening element. For this purpose, the catch strap mechanism can comprise one or more deflectors for the catch strap arrangement.

[0013] The invention further provides that two catch strap arrangements are provided, both of which are deflected by the one tightening element. It can be provided that both catch strap arrangements can be or are attached at different locations on the same airbag. However, it is preferred that both catch strap arrangements are attached with their ends to two airbags, in particular to different airbag modules. It is therefore possible to tighten two airbags with just one tightening device, so that the amount of equipment for tightening two airbags is reduced.

[0014] In one embodiment, it can be provided that the catch strap mechanism comprises a holder with which the ends of the catch strap arrangements that cannot be attached to the airbag are fixed. Depending on the arrangement of additional deflectors, it can therefore be achieved that the tightening path of the catch strap arrangements covered by the ends which can be attached to the airbag is more than 1.5 times, preferably 2 times or more than 2 times the distance covered by the tightening element.

[0015] In a further embodiment, it can be provided that the ends of the catch strap arrangements that are not to be attached to the airbags are fixed to the tightening element itself, wherein the catch strap arrangements are guided around a deflector between the other ends (fixed to the tightening element) and the portions deflected by the tightening element. The catch strap arrangements therefore extend from the tightening element through the deflector and back to the tightening element where they are deflected again. In this way, it is possible for the tightening path covered by the ends which can be attached to the airbag to be double or even more than double the tightening path covered by the tightening element.

Description

[0016] The invention and the technical environment are explained below by way of example with reference to the figures. Schematically, in the figures:

[0017] FIG. 1 is a perspective view of a vehicle seat with a catch strap mechanism for an airbag device in an initial state,

[0018] FIG. 2 shows the catch strap mechanism during a tightening process,

[0019] FIG. 3 shows a second embodiment of a catch strap mechanism in a vehicle seat in an initial state, and

[0020] FIG. 4 shows the catch strap mechanism according to FIG. 3 during a tightening process.

[0021] The figures show a catch strap mechanism in a vehicle seat 5. The catch strap mechanism comprises a first catch strap arrangement 1a which can be attached to an airbag by a first end 1a.1. The catch strap mechanism comprises a second catch strap arrangement 1b, which can be attached to another airbag by a first end 1b.1.

[0022] The catch strap mechanism also comprises a tightening device 2, which comprises a tensioner tube 2.3 extending in the longitudinal direction of a vehicle, to which a deflection block 2.4 is connected. The deflection block 2.4 has a receptacle 2.5 for a gas generator.

[0023] A piston (not visible in the figures) is arranged in the tensioner tube 2.3 and is connected to a tension cable 2.2. The tension cable 2.2 is led out of the tensioner tube 2.3 through the deflection block 2.4.

[0024] A tightening element 2.1 is attached to the tension cable 2.2, which deflects the catch strap arrangements 1a and 1b.

[0025] In the embodiment shown in FIGS. 1 and 2, both catch strap arrangements 1a and 1b are fixed to a holder 4 by their second ends 1a.2 and 1b.2.

[0026] If the tightening device is triggered, the piston arranged in the tensioner tube 2.3 is driven in the longitudinal direction of the vehicle so that the tightening element 2.1 in FIGS. 1 and 2 is accelerated in a straight line to the left. This movement of the tightening element 2.1 causes the ends 1a.1 and 1b.1 attached to the airbag to be retracted so that the airbag (not shown) is aligned after or during its unfolding. Since a type of pulley is formed by the deflection of the catch strap arrangements 1a and 1b by means of the tightening element 2.1, the tightening path covered by the ends 1a.1 and 1b.1 is greater than the tightening path covered by the tightening element 2.1.

[0027] In the embodiment shown in FIGS. 3 and 4, the ends 1a.2 and 1b.2 of the catch strap arrangements 1a and 1b are fixed to the tightening element 2.1. Both catch strap arrangements 1a and 1b are then deflected around a deflector 3 and by the tightening element 2.1.

[0028] During the tightening movement indicated in FIG. 4, the path covered by the ends 1a.1 and 1b.1 is again increased in comparison to the path covered by the tightening element 2.1.

[0029] By using a tension cable 2.2, the tensioner tube 2.3 can be aligned relatively freely to the linearly accelerated tightening element 2.1 and its linear tightening path. In addition, a tightening movement of two catch strap arrangements can be initiated with only one tightening element 2.1.

List of reference signs

[0030] 1a First catch strap arrangement [0031] 1b Second catch strap arrangement [0032] 1a.1 First end of the first catch strap arrangement [0033] 1b.1 First end of the second catch strap arrangement [0034] 1a.2 Second end of the first catch strap arrangement [0035] 1b.2 Second end of the second catch strap arrangement [0036] 2 Tightening device [0037] 2.1 Tightening element [0038] 2.2 Tension cable [0039] 2.3 Tensioner tube [0040] 2.4 Deflection block [0041] 2.5 Receptacle [0042] 3 Deflector [0043] 4 Holder [0044] 5 Vehicle seat

Claims

1. A catch strap mechanism for an airbag module having an airbag, comprising: at least one catch strap arrangement, wherein one end of a catch strap arrangement can be attached to the airbag, and a tightening device for the catch strap arrangement, wherein the tightening device comprises a tightening element which can be accelerated in a straight line, in particular pyrotechnically, wherein the tightening element deflects the catch strap arrangement, and two catch strap arrangements are provided, wherein the one tightening element deflects both catch strap arrangements.
2. The catch strap mechanism according to claim 1, wherein other ends of both catch strap arrangements are fixed to the tightening element, wherein the catch strap arrangements are guided around a deflector between the other ends and the portions deflected by the tightening element.

3. The catch strap mechanism according to claim 1, which comprises a holder with which the other ends of the catch strap arrangements can be fixed.
 4. The catch strap mechanism according to claim 1, wherein the tightening device comprises a tension cable by means of which the tightening element can be driven.
 5. The catch strap mechanism according to claim 1, wherein the tightening device comprises a piston connected to a tension cable, wherein the piston is arranged in a tensioner tube and can be driven by compressed gas which can be generated by a gas generator.
 6. The catch strap mechanism according to claim 1, wherein the tension cable is guided through a deflection block in which a receptacle for a gas generator is formed.
 7. An airbag device for a vehicle seat, comprising: at least two airbag modules, wherein each airbag module comprises at least one airbag and is configured for attachment to the vehicle seat, and the at least one catch strap mechanism according to claim 1, wherein the ends of the catch strap arrangements are respectively attached to an airbag.
 8. The airbag device according to claim 1, wherein the two airbag modules can be arranged on opposite sides of the backrest.
 9. A vehicle seat for a motor vehicle, comprising: a backrest, a seat cushion, and the catch strap mechanism for an airbag module having an airbag according to claim 1.
-